上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 3

Carbon Accounting: How

Haishan Yu

饮 水 思 源 · 爱 国 荣 校



How many greenhouse gases have been identified by physical scientists?

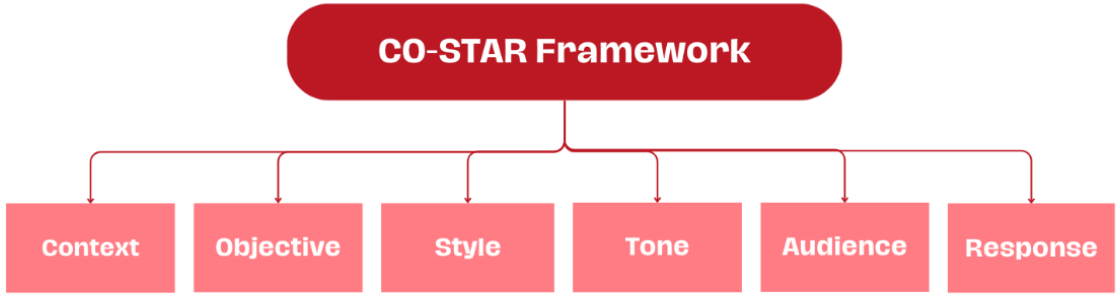


Prompt:

I am an undergraduate new to carbon accounting. My teacher asked, “Approximately how many greenhouse gases have been identified by physical scientists?”. Give one scientifically grounded approximate total from a source that directly states the number; do not infer it. Use plain, objective, and concise English in a neutral tone suitable for an undergraduate. Respond in one line only using this format: "Approximately X gases (counting criterion; source: one direct link)“.

The link must lead directly to a publicly accessible source, with no login, paywall, regional restriction, URL shortener, or search-result page.

CO-STAR Framework Breakdown

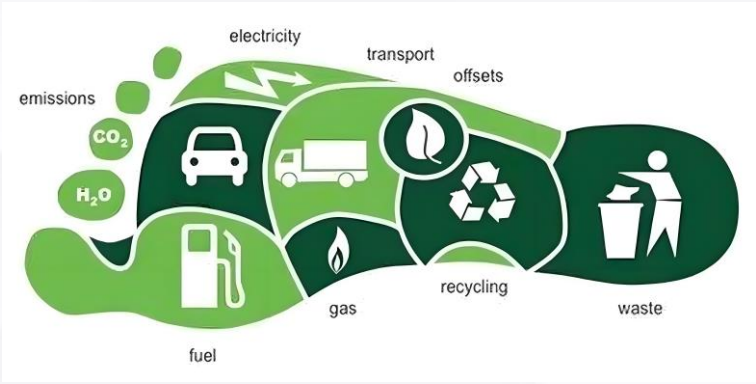


CO-STAR framework - Image by author

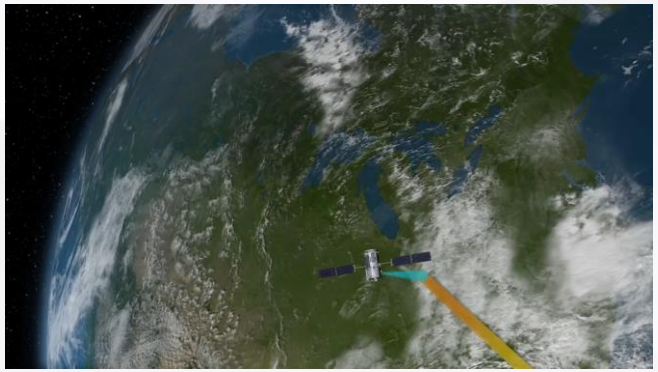
https://towardsdatascience.com/how-i-won-singapores-gpt-4-prompt-engineering-competition-34c195a93d41/

CO-STAR element	Prompt component
Context	*My teacher asked, "Approximately how many greenhouse gases have scientists identified?"*
Objective	*Give one scientifically grounded approximate total from a source that directly states the number; do not infer it.*
Style	*Use plain, objective, and concise English*
Tone	*Use a neutral tone.*
Audience	*Suitable for an undergraduate*
Response	*Respond in one line only: "Approximately X gases (counting criterion; source: one direct accessible link)."*

A Student's Carbon Footprint



Satellite inversion



CEMS / direct measurement



Li Ming



?



SO, HOW DO WE CALCULATE IT?

Learning Objectives

1. **Identify** the main greenhouse gases commonly covered in carbon accounting
2. **Explain** why different greenhouse gases are expressed in CO_2e and how GWP supports this conversion.
3. **Describe** how activities, data and emission factors are linked to GHG emissions.
4. **Develop** an initial activity–input–result map for the course project..

目 录

Contents

01-Fundamental Concepts and Quantification Formula

GHGs, GWP, CO_2e , basic conversions and formula

02-Accounting Guidelines

where accounting rules, methods come from

03-Wrap-up and next time

* In-class Project activity

Meet client, Complete the profile and develop the initial design brief

01

Fundamental Concepts and Quantification Formula

GHGs, GWP, CO_2e , basic conversions and formula

Greenhouse Gases in Carbon Accounting

Gas	Typical Sources	Practical Sensitivity
CO_2	Fossil fuel combustion, cement, land-use change	Usually dominant in energy-intensive systems
CH_4	Livestock, rice cultivation, landfills, fossil fuel leakage	Critical in agriculture, waste, and natural gas systems
N_2O	Fertilizer use, manure management, combustion, industry	Often important in agriculture and some industrial processes
HFCs / PFCs / SF_6 / NF_3	Refrigeration, electronics, power equipment, semiconductors	Small mass, potentially very high GWP

CO_2e: A Common Carbon Accounting Unit

- **CO_2 equivalent (CO_2e)** expresses the climate effect of different GHGs in a common unit.

$$Emissions\ in\ CO_2e = \sum_i Mass\ of\ GHG_i \times GWP_i$$

$CH_4 \rightarrow CO_2e$

$N_2O \rightarrow CO_2e$

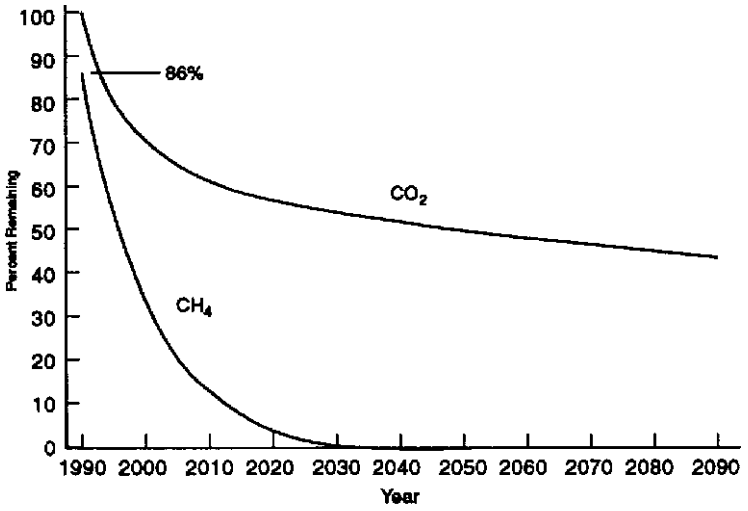
- What is GWP?
  - Global warming potentials (GWP) compares the **integrated warming effect** of a greenhouse gas with that of CO_2 over **a selected time horizon**.
    - Carbon dioxide (CO_2) always has a GWP of 1. All other greenhouse gases are measured/indexed relative to carbon dioxide.

$$GWP_i \equiv \frac{\int_0^{TH} RF_i(t)\,dt}{\int_0^{TH} RF_r(t)\,dt} = \frac{\int_0^{TH} a_i \cdot [C_i(t)]\,dt}{\int_0^{TH} a_r \cdot [C_r(t)]\,dt}$$

The formula is adapted from page 210 of IPCC (2007).

Why Does GWP Have Different Values?

- GWP values differ because of:
  1. Gas properties
     - Radiative efficiency
     - Atmospheric lifetime
  2. Time horizon
     - GWP20 emphasizes short-term warming
     - GWP100 is widely used in inventories and reporting



Scientific assessment regularly updates GWP values

- IPCC assessment reports update values as science improves

Table 7.15 | Emissions metrics for selected species: global warming potential (GWP), global temperature-change potential (GTP). All values include carbon cycle responses as described in Section 7.6.1.3. Combined GTPs (CGTPs) are shown only for species with a lifetime less than 20 years (Section 7.6.1.4). Note CGTP has units of years and is applied to a change in emissions rate rather than a change in emissions amount. The radiative efficiencies are as described in Section 7.3.2 and include tropospheric adjustments where assessed to be non-zero in Section 7.6.1.1. The climate response function is from Supplementary Material 7.SM.5.2. Uncertainty calculations are presented in Supplementary Tables 7.SM.8 to 7.SM.13. Chemical effects of CH_4 and N_2O are included (Section 7.6.1.3). Contributions from stratospheric ozone depletion to halogenated species metrics are not included. Supplementary Table 7.SM.7 presents the full table.

Species	Lifetime (Years)	Radiative Efficiency (W m^{-2} ppb^{-1})	GWP-20	GWP-100	GWP-500	GTP-50	GTP-100	CGTP-50 (years)	CGTP-100 (years)
CO_2	Multiple	$1.33 \pm 0.16 \times 10^{-5}$	1.	1.000	1.000	1.000	1.000		
CH_4-fossil	11.8 ± 1.8	$5.7 \pm 1.4 \times 10^{-4}$	82.5 ± 25.8	29.8 ± 11	10.0 ± 3.8	13.2 ± 6.1	7.5 ± 2.9	2823 ± 1060	3531 ± 1385
CH_4-non fossil	11.8 ± 1.8	$5.7 \pm 1.4 \times 10^{-4}$	79.7 ± 25.8	27.0 ± 11	7.2 ± 3.8	10.4 ± 6.1	4.7 ± 2.9	2675 ± 1057	3228 ± 1364
N_2O	109 ± 10	$2.8 \pm 1.1 \times 10^{-3}$	273 ± 118	273 ± 130	130 ± 64	290 ± 140	233 ± 110		
HFC-32	5.4 ± 1.1	$1.1 \pm 0.2 \times 10^{-1}$	2693 ± 842	771 ± 292	220 ± 87	181 ± 83	142 ± 51	78,175 ± 29,402	92,888 ± 36,534
HFC-134a	14.0 ± 2.8	$1.67 \pm 0.32 \times 10^{-1}$	4144 ± 1160	1526 ± 577	436 ± 173	733 ± 410	306 ± 119	146,670 ± 53,318	181,408 ± 71,365
CFC-11	52.0 ± 10.4	$2.91 \pm 0.65 \times 10^{-1}$	8321 ± 2419	6226 ± 2297	2093 ± 865	6351 ± 2342	3536 ± 1511		
PFC-14	50,000	$9.89 \pm 0.19 \times 10^{-2}$	5301 ± 1395	7380 ± 2430	10,587 ± 3692	7660 ± 2464	9055 ± 3128		

$$GWP_i \equiv \frac{\int_0^{TH} RF_i(t)\,dt}{\int_0^{TH} RF_r(t)\,dt} = \frac{\int_0^{TH} a_i \cdot [C_i(t)]\,dt}{\int_0^{TH} a_r \cdot [C_r(t)]\,dt}$$

IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change

Which GWP Should We Use?

- GWP in the standards

- UNFCCC national inventories now use AR5 (used SAR until 2015)
- GHG Protocol Standards – most up-to-date GWPs (adapted from AR5, recommended)
- ISO 14064-1:2018 – the latest IPCC's GWP should be used. If not, justification shall be provided.
- PAS 2050 - most up-to-date GWPs

Take away message: always check the requirements of the standard used.

C, CO_2, and CO_2e: Different Units, Different Meanings

Carbon (C): mass of carbon element
Carbon dioxide (CO_2): mass of carbon dioxide molecule
CO_2e: climate effect expressed as equivalent CO_2

C = 12

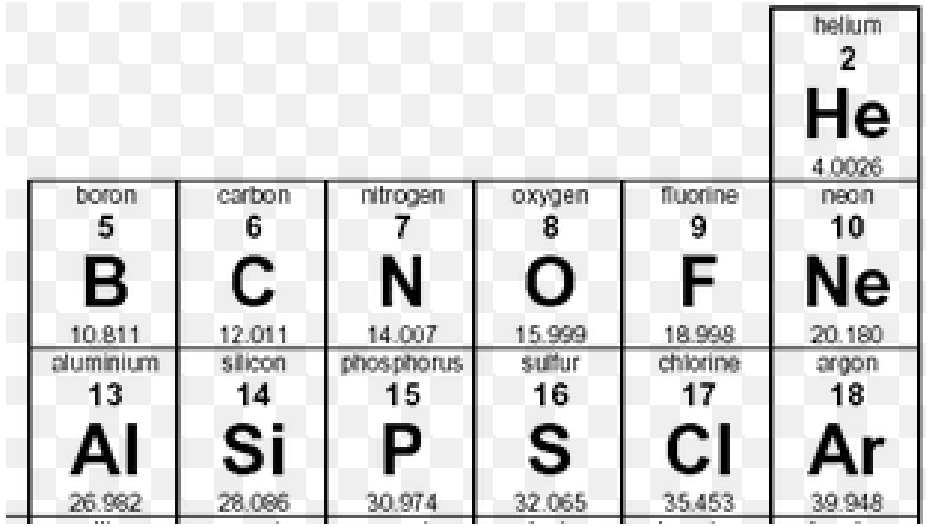
$$CO_2 = C \times \frac{44}{12}$$

CO_2 = 44

$$C = CO_2 \times \frac{12}{44}$$

1 tonne C = 3.667 tonnes CO_2

1 tonne CO_2 = 0.273 tonnes C



Quick quiz: Unit Conversion Practice

Q1. Chemical mass conversion

A dataset reports **12 tonnes C** emitted from fuel combustion. How many tonnes CO_2 is this?

Q2. GWP-based conversion

A landfill emits **2 tonnes CH_4**. If GWP100 for CH_4 is **27.2**, what is the emission in tonnes CO_2e?

Q3. Guideline choice

You find two GWP values for CH_4 in two references. Which one should be used in an official carbon accounting report?

A. The latest scientific value available online
B. The value specified by the applicable accounting guideline
C. The larger value to be conservative
D. The smaller value to avoid overestimation

Basic formula for Calculation-Based Accounting

- Core calculation structure:

$$Emissons = Activity\ Data\ \times Emission\ Factor$$

Activity data: amount of activity causing emissions

Emission factor: emissions per unit of activity

Boundary: what sources are included

Source and sink: where GHGs are emitted or removed

The formula is simple; the accounting judgment is not.

02

Carbon Accounting Standards

Why Do We Need Accounting Standards?

$$Emissons = Activity\ Data\ \times Emission\ Factor$$

Which activities and emission sources should be included?

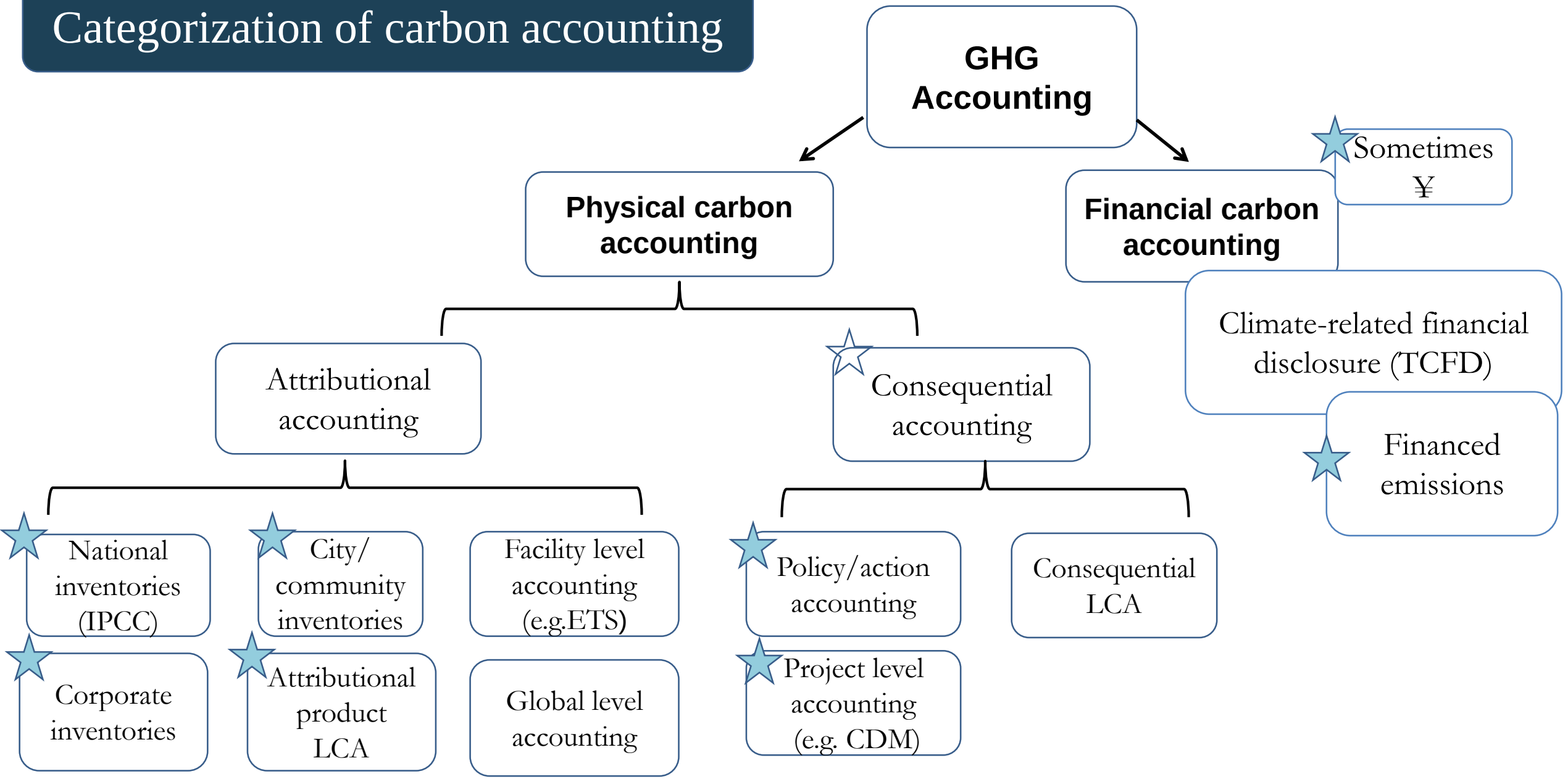
Which data and emission factors should be used?

How should different gases and missing data be treated?

What should be reported with the result?

Accounting standards turn a general calculation formula into a defined and reportable accounting practice.

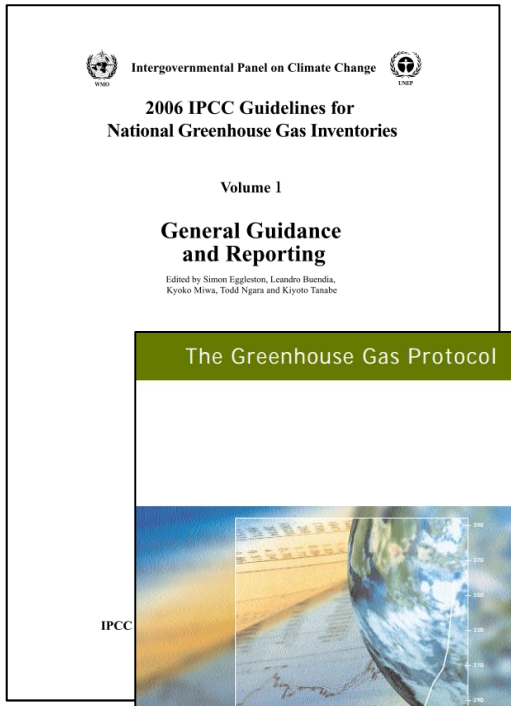
Categorization of carbon accounting

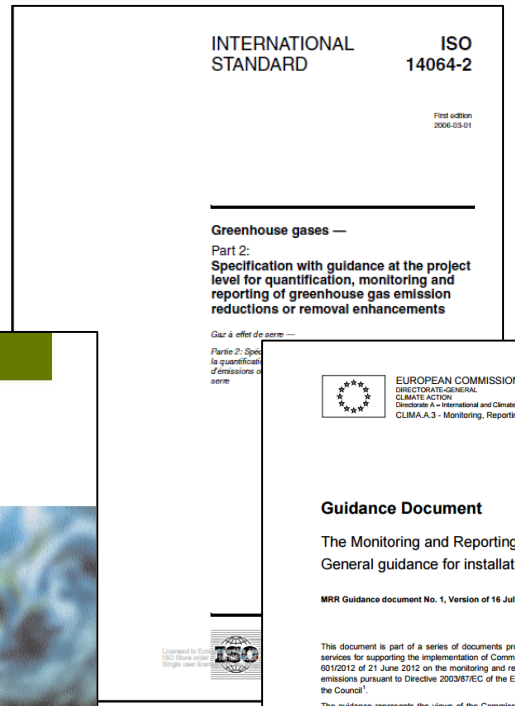


Different Accounting Applications, Different Guidelines

ACCOUNTING APPLICATION	EXAMPLES of STANDARDS
National GHG inventories	IPCC Guidelines
City or community inventories	GPC or applicable local protocol
Organizational inventories	GHG Protocol or ISO 14064-1
Product carbon footprints	ISO 14067 or a product carbon-footprint standard
Project or policy assessment	Project- or policy-level accounting guidance

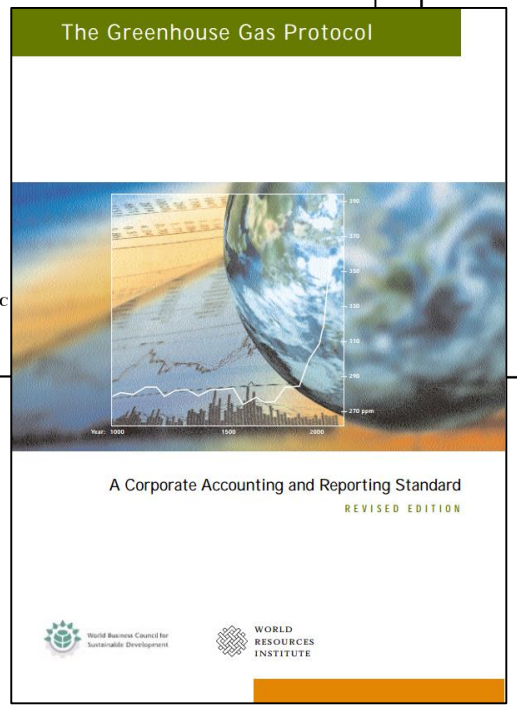
Many different international Carbon Accounting standards

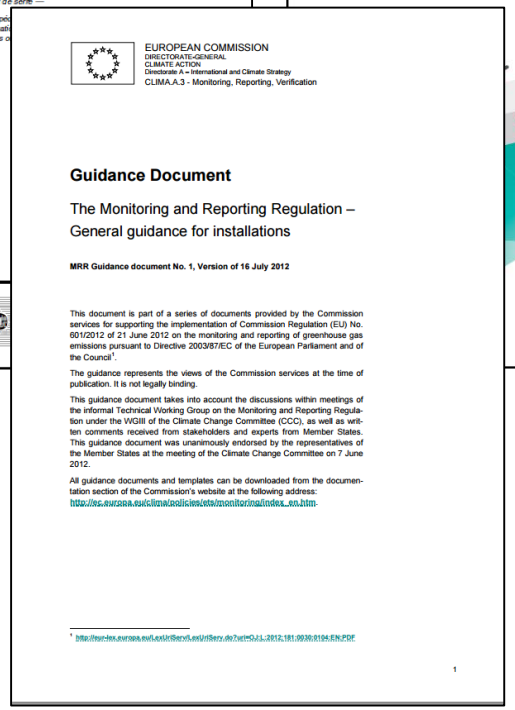


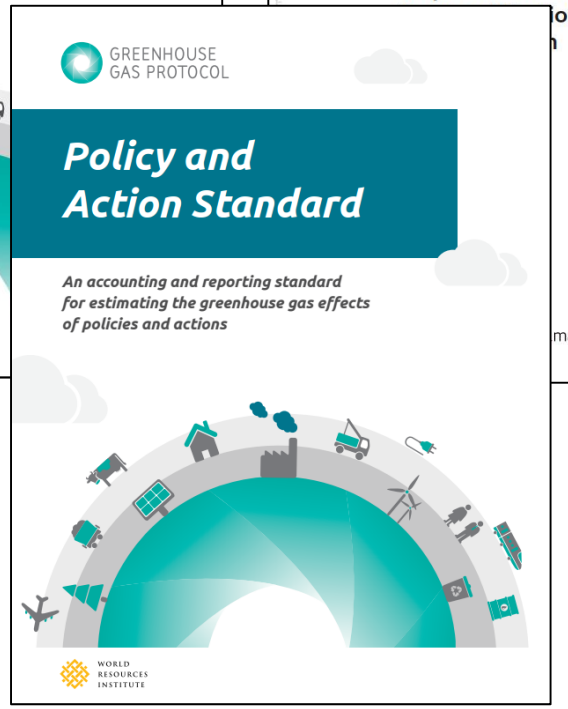














Examples of Guidelines in China

https://std.samr.gov.cn/

- **GB/T 32150-2025** General guidelines of the greenhouse gas emissions accounting and reporting for industrial enterprises (implemented from 2026.7.1)
  - GB/T 32151.1-2015 Requirements of the greenhouse gas emission accounting and reporting—Part 1: Power generation enterprise
  - GB/T 32151.2-2015 (Power grid enterprise)

    ...
  - GB/T 32151.54-2025 (Industrial sulphuric acid enterprise)
- **GB/T 24067-2024** Greenhouse gases—Carbon footprint of products—Requirements and guidelines for quantification
- Guidelines for the Compilation of **Provincial** Greenhouse Gas Inventories (2025 Edition)

上海节能低碳和应对气候变化网 (reg-sh.org) (unavailable at this moment)

上海市温室气体排放核算与报告指南（试行） （沪发改环资〔2012〕180 号)

上海市电力、热力生产业温室气体排放核算与报告方法（试行） （沪发改环资〔2012〕181 号)

钢铁行业

化工行业

有色金属行业

纺织和造纸行业

非金属矿物制品业

航空运输业

旅游饭店、商场、房地产业及金融业办公建筑

运输站点行业

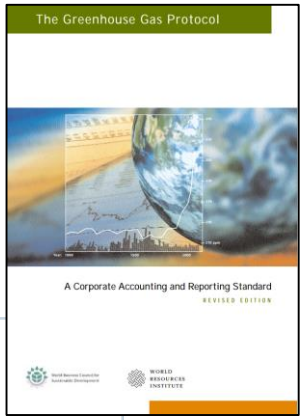
水运业

In Shanghai

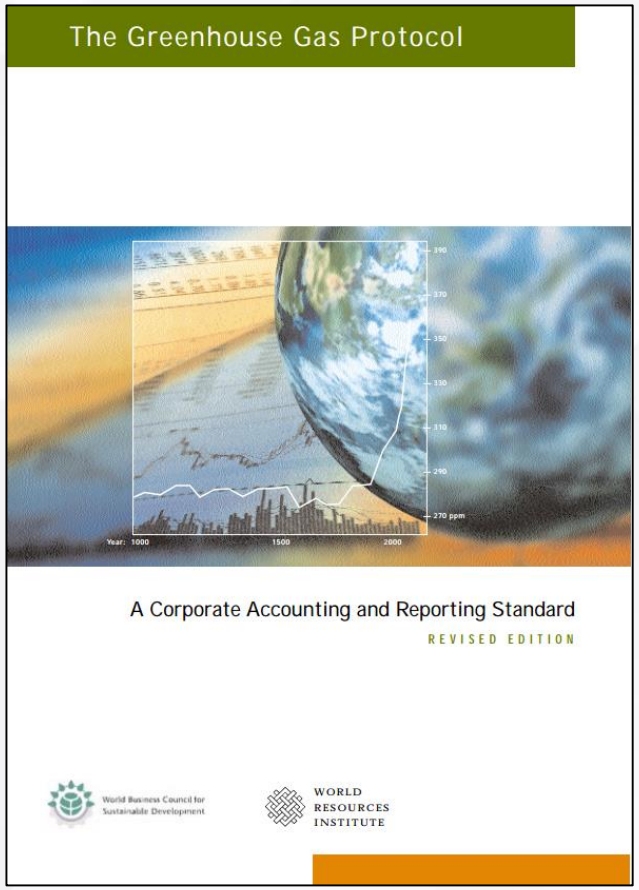


Inside an Accounting Standard

The **GHG Protocol Corporate Standard** as an example



Content	TYPICAL QUESTIONS ADDRESSED
Principles and inventory design	Accounting principles; business goals; inventory design
Boundaries and classification	Organizational boundaries; operational boundaries; scopes
Quantification and tracking	Emission sources; activity data; emission factors; calculations; base year and recalculation
Quality, reporting and verification	Inventory quality management; reporting requirements; verification
***Reductions and targets**	Accounting for reductions and offsets; setting and tracking target



Introduction	The Greenhouse Gas Protocol Initiative		2
Chapter 1	GHG Accounting and Reporting Principles	STANDARD GUIDANCE	6
Chapter 2	Business Goals and Inventory Design	GUIDANCE	10
Chapter 3	Setting Organizational Boundaries	STANDARD GUIDANCE	16
Chapter 4	Setting Operational Boundaries	STANDARD GUIDANCE	24
Chapter 5	Tracking Emissions Over Time	STANDARD GUIDANCE	34
Chapter 6	Identifying and Calculating GHG Emissions	GUIDANCE	40
Chapter 7	Managing Inventory Quality	GUIDANCE	48
Chapter 8	Accounting for GHG Reductions	GUIDANCE	58
Chapter 9	Reporting GHG Emissions	STANDARD GUIDANCE	62
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Appendix A	Accounting for Indirect Emissions from Electricity		86
Appendix B	Accounting for Sequestered Atmospheric Carbon		88
Appendix C	Overview of GHG Programs		90
Appendix D	Industry Sectors and Scopes		92

What Does This Guideline Actually Standardize?

- Principles — what are generally accepted carbon accounting principles;
- Boundary — which organizational activities enter the inventory;
- Classification — how emissions are assigned to scopes;
- Quantification — how sources, data and factors are used;
- Consistency and quality — how emissions are tracked and checked;
- *Setting the target and Reporting — what must be disclosed with the result.



Which standard should I use?

SYNTHESIS –
What Makes an Emissions Estimate Useful?



Image credit: NOAA

Context
Where is action or accountability needed?

→

User
Who must understand, act or answer?

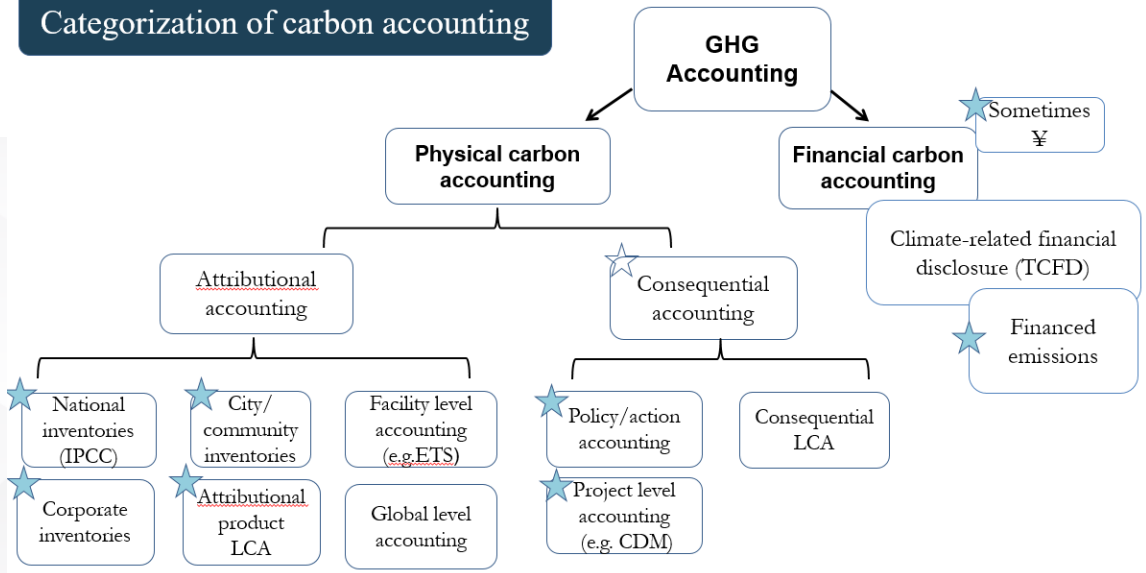
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Purpose
What question, claim or decision is at stake?

A carbon account becomes useful when these elements align.

A question we will return to: Is the accounting result fit for its intended use?

Categorization of carbon accounting





04

Wrap-up and Next time

Wrap up

What we have built today

1. Common measurement language
   - Different GHGs are converted into CO_2e using GWP
   - GWP value is scientific, and in carbon accounting it must follow the relevant guideline or reporting rule
   - C, CO_2, and CO_2e can not be mixed
2. Carbon accounting standards
   - Different objects require different boundaries, data, and guidelines
   - Guidelines specify boundaries, methods, factors and reporting—but choices still need explanation

Next: attributional vs consequential carbon accounting—and the common analytical framework.

Suggested Readings for Lecture 4

- Essential reading:
  - Brander, M. 2021. “The Most Important GHG Accounting Concept You May Not Have Heard of: The Attributional-Consequential Distinction.” a blog at https://ghginstitute.org/2021/04/21/the-most-important-ghg-accounting-concept-you-may-not-have-heard-of-the-attributional-consequential-distinction
- Optional reading
  - Russell, S. 2019. Estimating and Reporting the Comparative Emissions Impacts of Products. Access at https://ghgprotocol.org/sites/default/files/2023-03/18_WP_Comparative-Emissions_final.pdf
- **Further reading:**
  - https://ourworldindata.org/ghg-emissions-by-sector

In-class Course Project Activities

Mind map: the first design brief

COURSE PROJECT

Your Design Challenge

Build a web-based personal carbon footprint calculator that a real SJTU student can use.

WHAT THE STUDENT SHOULD BE ABLE TO DO

1. **Enter activities**
   Transport, meals, accommodation, energy consumption, and any other activities of interest
2. **Calculate emissions**
   Computes emissions from the entered activities, reported in kg or tonnes of CO_2e.
3. **Explore the results**
   View the total, browse by category, and compare across activities or against averages.
4. **Get reduction tips**
   Actionable suggestions for lowering emissions, tied to each activity.

YOUR GROUP DECIDES

- **What the tool asks, includes, calculates, and explains**

AI can help build the webpage
— but your group makes and explains the carbon-accounting choices behind the calculator.

PRODUCT REQUIREMENTS

What Will Matter Most?

The project is assessed as a carbon-accounting tool—not as a software competition.

Reasonable

The design fits the intended user and purpose.

Correct

Calculations, units and interpretation are sound and defensible.

Transparent

Boundaries, assumptions, emission factors and limitations are clearly disclosed.

Usable

An SJTU student can enter the required information and obtain clear results that support their decision-making.

A polished interface may improve the experience, but technical complexity is not the deciding criterion.

PROJECT MILESTONES

Course Project Timeline

TODAY

1

Client brief

Shared facts, group assumptions and a mind map

LECTURE 4

2

Prototype 0

First runnable version and the original AI prompt

MID-COURSE

3

Version 1

Test with new accounting concepts and explain revisions

LATER MODULES

4

Version 2

Challenge earlier choices and extend the tests

END OF COURSE

5

Final product

Final tool, version record and project presentation

Keep every version and its prompt log.

Record why the group changes the accounting design.

TODAY'S GROUP TASK – "Vibe"

Meet Li Ming -- and Create the first design brief by drawing a mind map

SHARED PROFILE

Begin with the same factual profile on the task sheet (see Canvas).

+

GROUP ADDITIONS

Add plausible characteristics that you want to enrich the figure. Label them clearly as group assumptions.

Before building a calculator, describe the student and the experience the tool should create.

Build your mind map around six anchors:

- Add characteristics that may affect how the tool should be designed. (**CLIENT**)
- State what the calculator should help Li Ming understand or do. (**PURPOSE**)
- Identify the activities the calculator might include. (**ACTIVITIES**)
- Decide what information Li Ming may need to enter. (**INPUTS**)
- Show how one or two inputs could lead to emissions estimates. (**LOGIC**)
- Decide what results and explanations the calculator should provide. (**OUTPUTS**)

WORKING RULES

- Focus on ideas, not artistic quality.
- Use keywords, short notes and arrows
- Mark group assumptions and open questions.
- Make the map readable to everyone.
- Submit the paper after class.

Keep the map. Next class, it becomes the brief for your first AI prompt.





See you next time